

MTU_ValueService **Technical Documentation**

CAN-Bus-Interface

MTU-MDEC to third-party systems

MTU/DDC Series 4000

DDC/MTU Series 2000

Stationary genset applications

Documentation Part 1

- Structure and function

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Printed in Germany

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Salvo alterações.

General information about documentation

Documentation structure

Documentation Part	Title/contents	Target group(s)
1	Structure and function	Operating personnel etc.
2	Operation	Operating personnel
3	Maintenance and repair	Plant personnel
4	Service manual	Electronic service personnel
5	Illustrated parts catalog	Plant personnel, electronic service personnel, logistics personnel
6	Plant-specific configuration	Electronic service personnel, installation personnel, start-up personnel
7	Mechanical and electrical installation	Installation personnel (electromechanical/mechatronic engineer)
8	Initial start-up	Start-up personnel

Note: Not all parts of the documentation are written for every product.

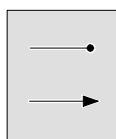
Required knowledge

To understand each part of the documentation, we recommend reading the preceding parts, if applicable.

Reference numbers and reference lines

Details in figures are provided with reference numbers and reference lines if necessary.

If reference is made in the text to a detail provided with a reference number, the figure number and, separated by an oblique, the reference number of the detail are written in brackets. Example: (5/2) means fig. 5, reference number 2.

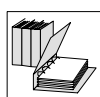


A point at the end of the reference line means that the detail is visible in the figure.

An arrow at the end of the reference line indicates that the detail cannot be seen in the figure.

Symbols

The symbols used in safety notes are defined in the chapter "Safety requirements".



This symbol indicates cross-references to other manuals.



This symbol indicates important information.



This symbol indicates detailed information.



Further troubleshooting or fault rectification of a fault marked with this symbol requires mechanical work to be performed on other assemblies or equipment with reference to the relevant documentation.



Further troubleshooting or fault rectification of a fault marked with this symbol requires work to be performed on the engine with reference to the engine documentation.



Troubleshooting or rectification of a fault marked with this symbol either requires the customer's trained service personnel or service personnel provided by MTU. The affected assembly may have to be sent to MTU for repairs. Such a fault cannot be repaired by plant personnel.

Some of the chapters in this manual contain structured task descriptions. The symbols used here are explained at the relevant points by associated sub-titles.

Part 1

Structure and function

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1 General information

Please note the following general information:

- MTU uses the Basic CAN 2.0A in accordance with the ISO 11898.
- The baud rate is 125kBaund.
- The bus is a two-wire connection with 120 ohm terminators at the ends of the CAN bus connection.
- Initialization values of the CAN control block:

btr0:	0X43
btr1:	0X1C

The input parameter 200.00 ("CAN Interface Config Param") must be defined by the customer and MTU prior to initial start-up. This input parameter makes the customer-specific settings of the ECU (e.g. whether one CAN bus connection is active or two CAN bus connections are active).

1.1 System overview

This chapter provides an overview of the structure of

- The entire MCS-5 system
- The field bus 1 communication protocol

1.1.1 Structure of the MCS-5 system

In the MTU automization system "Monitoring and Control System MCS-5" a differentiation is made between:

- Management level
- Process level

A maximum of 31 devices may be connected to field bus 1 in the MCS-5 system (1 PPS + 30 MTU I/O devices or external third-party devices). The figure below illustrates the structure of an MCS-5 system.

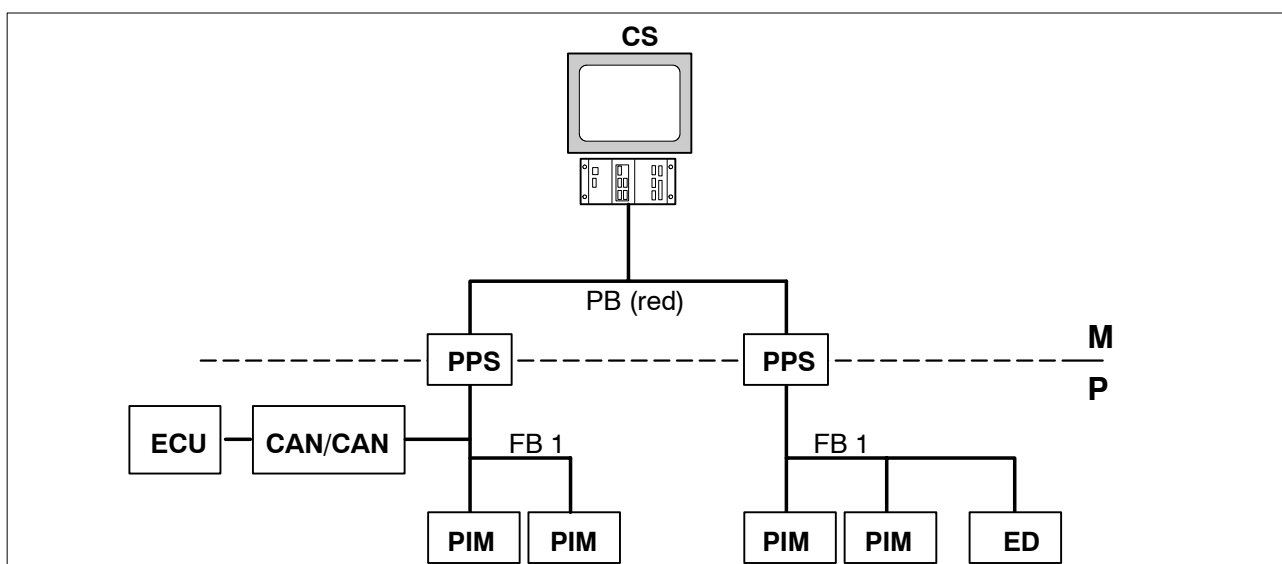


Fig. 1 : Structure of an MCS-5 system

PB (red)	Process bus, redundant	FB	Field bus 1
PPS	Programmable Process Station	CAN/CAN	Coupler
PIM	I/O module (MTU)	ED	External third-party device
ECU	Engine Control Unit	CS	Control station
M	Management level	P	Process level

1.1.2 Structure of the communication protocol in layer 1

A dedicated communication protocol has been developed for the MCS-5 system. It corresponds to the ISO/OSI layer model whereby only layers 1, 2 and 7 are used. Parts of the other layers are incorporated in layer 7.

Standard CAN frames are used exclusively in communication layer 1. The CAN telegram for layer 1 is as follows:

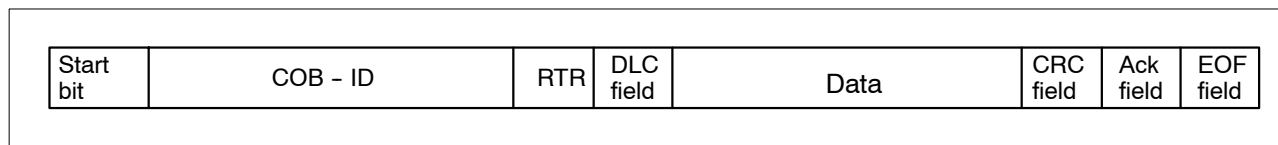


Fig. 2 : Structure of the CAN telegram in layer 1

Certain parts of this telegram, particularly the data field (field 5), are described in more detail in this manual.

1.1.2.1 Brief description of the attributes in the communication protocol

Start bit	Start bit (dominant bit, identifies the beginning of the telegram)
COB-ID	CAN Object Identifier (11 bits)
RTR	Remote Transmission Request (1 bit) Always set to zero.
DLC	Data Length Code field Length (in bytes) of useful data ("data field"); max. adjustable length: 8 bytes
Data field	Useful data The length of the data field is: 0 to 8 bytes
CRC	Test sequence to detect data transmission errors, length: 15 bit CRC + 1 bit CRC delimiter
Ack field	Ack field Used for positive acknowledgement of receipt by each CAN bus node regardless of station.
EOF	End Of Frame Every data telegram and data request telegram terminates in a bit sequence of 7 recessive bits.